

Analog Mixing Console with Digital Control and Automated Signal Path Configuration

Capstone Project Report

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October 9, 2026

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1 Introduction

1.1 Background

1.2 Problem Statement

1.3 Scope and Design Goals

2 Analog Design

2.1 Analog Systems Overview

2.2 Voltage Controlled Amplifiers

The most fundamental part of an audio mixer is control of each individual channels' gain before summing to the final stereo signal. To implement gain control that can be remote controlled by a voltage, voltage controlled amplifiers (VCAs) were used. Two VCAs were actually used on every channel. Since we aim to build a stereo mixer, the two VCAs feed the left and right summing channel of the mix bus, respectively. Each channel's two VCAs may be given different gain control voltages (CVs) to pan the channel's audio in the stereo field. For example, if a channel's left VCA had a CV to attenuate by -90dB (basically muted) and the right VCA had the CV to add 6dB of gain, that channel's audio would sound like it was coming from the right speaker after summing. To implement the regular fader behavior, which changes the gain of the left and right side together, both the left and right control voltages would be scaled together to change gain equally in both channels.

The specific VCA we picked for this purpose was the THAT 2180 from the Massachusetts-based THAT Corporation. It is the industry standard VCA for high fidelity audio performance. This is due to its total harmonic distortion (THD) of just 0.005% (for the 'A' grade 2180s), which is extremely low for a VCA. Additionally, it uses a Blackmer gain cell, which has a logarithmic gain curve, allowing control voltage and decibel gain to coincide linearly. Unfortunately, in September 2026, halfway through the design of the project, THAT Corporation announced it was discontinuing its legendary 2180 series, sending shockwaves through the audio electronics community. Frantically, we bought 32 of the 'A' grade THAT 2180s to use for this purpose, but the rest of the analog systems use clones of the THAT 2180 (Alfa RPAR AS2180, which from our tests, seems to have nearly as good performance as the genuine 'A' grade 2180s) or the SSI2164 VCA.

Placeholder: one VCA channel — C_{in} , R_{in} , THAT2180A, I/V converter
 $(R_f \parallel C_f)$, control-port snubber, R_{SET}

Figure 1: One channel of the pan/fader VCA stage.

Current-mode operation.

$$I_{\text{in}} = \frac{V_{\text{in}}}{R_{\text{in}}}, \quad I_{\text{out}} = A_I I_{\text{in}}, \quad V_{\text{out}} = R_f I_{\text{out}} \implies \frac{V_{\text{out}}}{V_{\text{in}}} = \frac{R_f}{R_{\text{in}}} A_I \quad (1)$$

Exponential gain control.

$$A_I = \exp\left(\frac{q(E_{C+} - E_{C-})}{2kT}\right) \implies G_{\text{dB}} = 20 \log_{10} A_I = \frac{E_{C+} - E_{C-}}{\kappa(T)}, \quad (2)$$

$$\kappa(T) = \frac{\ln 10}{10} \frac{kT}{q} \approx 6.1 \text{ mV/dB}$$

Temperature sensitivity.

$$G_{\text{dB}}(T) = G_{\text{dB}}(T_0) \frac{T_0}{T} \implies \frac{\Delta G_{\text{dB}}}{G_{\text{dB}}} \approx -\frac{\Delta T}{T_0} \approx -0.33 \%/\text{K} \quad (3)$$

Operating point.

$$I_{\text{SET}} = \frac{V_{\text{pin5}} - V_{EE}}{R_{\text{SET}}} = \frac{-2.85 \text{ V} + 15 \text{ V}}{5.1 \text{ k}\Omega} = 2.38 \text{ mA} \geq \hat{I}_{\text{in}} + \hat{I}_{\text{out}} + 570 \mu\text{A} = 1.97 \text{ mA} \quad (4)$$

Component	Value	Purpose
R_{in}, R_f	20 k Ω	unity gain at 0 dB; datasheet noise/THD point
C_{in}	10 μF bipolar	$f_c = 1/(2\pi R_{\text{in}} C_{\text{in}}) = 0.8 \text{ Hz}$; blocks DC
C_f	22 pF	cancels 15 pF at pin 8 (I/V stability)
R_{SET}	5.1 k Ω	$I_{\text{SET}} = 2.38 \text{ mA}$, Eq. (4)
Port snubber	100 Ω + 2.2 nF to ground	damps inductive op-amp output
Sym (pin 4)	open	keeps factory THD trim

Table 1: THAT2180A operating point.

Pan and fader law.

$$g_L = G \cos\left(\frac{\pi\theta}{2}\right), \quad g_R = G \sin\left(\frac{\pi\theta}{2}\right), \quad g_L^2 + g_R^2 = G^2, \quad \theta \in [0, 1] \quad (5)$$

$$G_{\text{dB},L} = 20 \log_{10} G + 20 \log_{10} \cos\left(\frac{\pi\theta}{2}\right), \quad G_{\text{dB},R} = 20 \log_{10} G + 20 \log_{10} \sin\left(\frac{\pi\theta}{2}\right) \quad (6)$$

2.3 Control Voltage Buffering and Temperature Drift Compensation

Placeholder: CV buffer — DAC input, R_{cv} + trim, R_{off} to -2.5 V,
 $R_p \parallel R_{NTC}$, MFB (R_2 , R_3 , C_1 , C_2), OPA2197, snubber at E_{C+}

Figure 2: Control voltage buffer for one VCA.

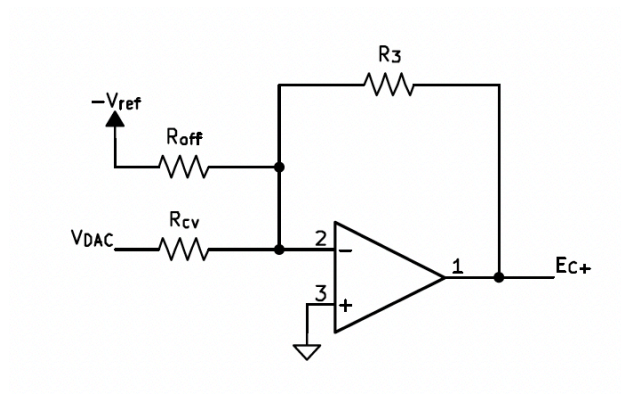


Figure 3: TEST

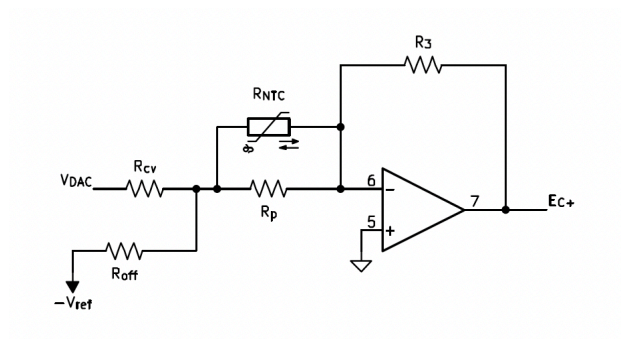


Figure 4: TEST

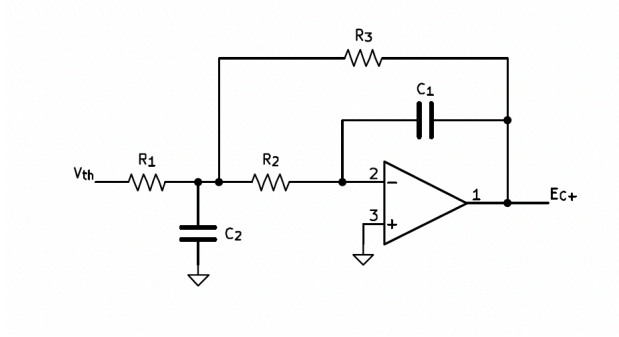


Figure 5: TEST

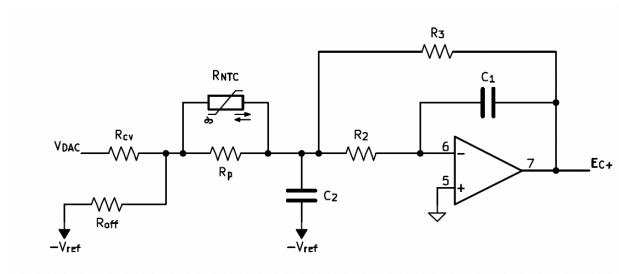


Figure 6: TEST

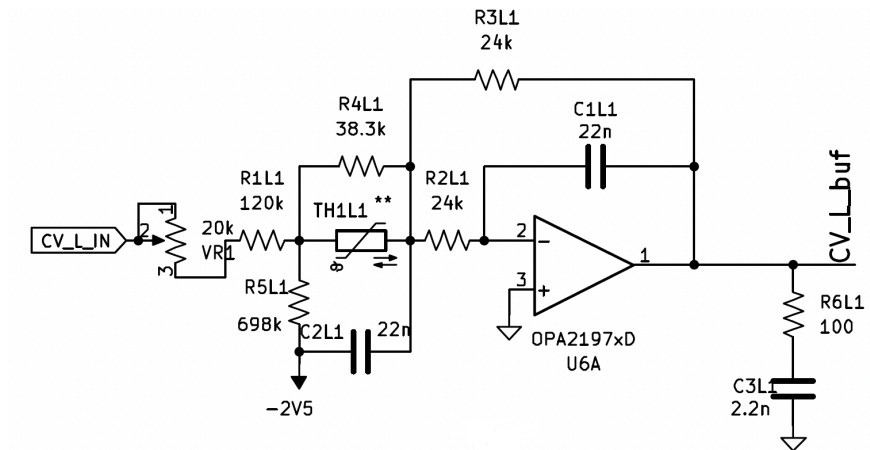


Figure 7: The trim potentiometer on the left is ONLY to calibrate temperature scaling during assembly, and it is not accessible to the end user.

Component	Value	Role
R_{cv}	120 k Ω + 20 k Ω trim (133 k Ω nom.)	DAC voltage to current; span trim
R_{off}	698 k Ω	offset current
$R_p \parallel R_{NTC}$	38.3 k $\Omega \parallel 47$ k Ω ($B = 4090$ K)	temperature compensation
$R_2 = R_3$	24 k Ω	filter and DC gain
$C_1 = C_2$	22 nF film	filter
V_{ref}	2.5 V (ADR5041 at -2.5 V, 222 μ A bias)	offset reference
Op-amp	OPA2197	low offset and noise

Table 2: CV buffer component values (one channel).

DAC.

$$V_{DAC} = \frac{2V_{REF}}{4096} n = n \times 1 \text{ mV}, \quad V_{REF} = 2.048 \text{ V}, \quad n \in \{0, \dots, 4095\} \quad (7)$$

DC transfer.

$$R_{th} = R_{cv} \parallel R_{off}, \quad V_{th} = R_{th} \left(\frac{V_{DAC}}{R_{cv}} - \frac{V_{ref}}{R_{off}} \right), \quad Z_t = R_p \parallel R_{NTC}(T), \quad R_1 = R_{th} + Z_t \quad (8)$$

$$E_{C+} = -\frac{R_3}{R_1} V_{th} = -R_3 D \left(\frac{V_{DAC}}{R_{cv}} - \frac{V_{ref}}{R_{off}} \right), \quad D = \frac{R_{th}}{R_{th} + Z_t} \quad (9)$$

$$E_{C+} = 72.3 \text{ mV} - 0.1518 V_{DAC} \implies G_{dB} = 11.9 \text{ dB} - 24.9 \frac{\text{dB}}{\text{V}} V_{DAC} \quad (10)$$

Code mapping.

$$n = \frac{72.3 \text{ mV} - \kappa G_{dB}}{0.1518 \times 1 \text{ mV}} \quad (11)$$

G_{dB}	+11.9	+6	0	-3.01	-20	-70	-90
n	0	235	476	597	1280	3290	4095

Table 3: Example codes from Eq. (11) (nominal, before calibration).

Filtering.

$$\frac{E_{C+}(s)}{V_{th}(s)} = -\frac{R_3}{R_1} \frac{1}{1 + sC_1 \left(R_2 + R_3 + \frac{R_2 R_3}{R_1} \right) + s^2 R_2 R_3 C_1 C_2} \quad (12)$$

$$f_0 = \frac{1}{2\pi \sqrt{R_2 R_3 C_1 C_2}} = 301 \text{ Hz}, \quad Q = \frac{\sqrt{R_2 R_3 C_1 C_2}}{C_1 (R_2 + R_3 + R_2 R_3 / R_1)} = 0.46 \quad (13)$$

Temperature compensation.

$$E_{C+}(T) = E_{C+}(T_0) \frac{T}{T_0} \implies \frac{1}{E_{C+}} \frac{dE_{C+}}{dT} = \frac{1}{T_0} \approx 0.33\%/K \quad (14)$$

$$\frac{1}{D} \frac{dD}{dT} = -(1-D) \frac{1}{Z_t} \frac{dZ_t}{dT} \quad (15)$$

$$R_{\text{NTC}}(T) = R_{25} e^{B\left(\frac{1}{T} - \frac{1}{T_{25}}\right)}, \quad \frac{dR_{\text{NTC}}}{dT} = -\frac{B}{T^2} R_{\text{NTC}}, \quad \frac{dZ_t}{dT} = \left(\frac{R_p}{R_p + R_{\text{NTC}}}\right)^2 \frac{dR_{\text{NTC}}}{dT} \quad (16)$$

$$(1-D) \left| \frac{1}{Z_t} \frac{dZ_t}{dT} \right|_{T_0} = \frac{1}{T_0} \quad (17)$$

$$\begin{aligned} \left. \frac{dR_{\text{NTC}}}{dT} \right|_{T_0} &= -2.16 \text{ k}\Omega/\text{K}, & \frac{1}{Z_t} \frac{dZ_t}{dT} &= \frac{-436 \Omega/\text{K}}{21.1 \text{ k}\Omega} = -2.07\%/K, \\ 1-D &= 0.159, & \frac{1}{D} \frac{dD}{dT} &= 0.159 \times 2.07\%/K = 0.33\%/K \end{aligned} \quad (18)$$

Calibration.

$$n(G_{\text{dB}}) = n_a + (G_{\text{dB}} - G_a) \frac{n_b - n_a}{G_b - G_a} \quad (19)$$

2.4 Channel Routing and Auxiliary Buses**2.5 Voltage Controlled State Variable Filters and EQ Circuit****2.6 Input Circuitry and Preamplifier****2.7 Hybrid Digital/Analog Dynamic Range Compressor****2.8 Mix Bus and Master Section****2.9 Channel Insert Routing Matrix****2.10 PCB Design Notes**

3 Digital Systems

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4 Testing and Results

4.1 Analog Performance Metrics

4.1.1 Linearity / Distortion

4.1.2 Noise

4.1.3 Temperature Drift Compensation

4.2 Digital System Bug Testing

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